SMKS-1

VL 1 - Grundlegende Prinzipien

Lambert-Beer: $E = \log \frac{I_0}{I} = \varepsilon c d = \sigma N d$

Absorptionsquerschnitt: $\sigma = \frac{\varepsilon}{N_A} \ln(10)$

$I = \frac{E_0^2}{2\mu_0 c}$ $\mu_0 = \frac{1}{\varepsilon_0 c^2} \to E_0 = \sqrt{\frac{2I}{\varepsilon_0 c}}$

$B_0 = \frac{2I/\mu_0}{E_0} = \frac{2I}{E_0 \varepsilon_0 c^2}$ $E = h\nu$

$I = \frac{\dot{n} h\nu}{A} \to \dot{n} = \frac{IA}{h\nu} = \frac{I A \lambda}{hc}$

Leistung, Energiedichte, Leistungsdichte

$P = \nu c A$

$F = \nu A = \frac{P}{c}$ $I = \frac{P}{A}$

$\nu = n_{ph} \cdot E_{ph}$ $n_{ph} = \frac{\nu_{ph}}{V}$

$V = A \cdot x$ $P_{ph} = \frac{h}{\lambda}$

$v = \frac{P}{m} = \frac{h \nu_{ph}}{m\lambda} = \frac{h E_{ges}}{m \lambda E_{ph}} = \frac{E_{ges}}{mc} = \frac{Pt}{mc}$

VL 2 - Resonanz

Gedämpfter harm. Oszillator

Schwingung $x(t) = x_0 e^{-\beta t} \cos(\omega_s t)$

Einhüllende $x(t) = x_0 e^{-\beta t}$

Schwache Dämpfung $\omega_s = \sqrt{\omega_0^2 - \beta^2}$ mit $\omega_0 = \sqrt{\frac{k}{m}}$ (Dämpfung, Federkonstante; oft $\omega_s = \omega_0$)

$\Rightarrow x_0$ - Amplitude, $\omega = \frac{2\pi}{T}$ - Kreisfrequenz, $\beta = \frac{-\ln\left(\frac{x(t)}{x_0}\right)}{t}$ - Dämpfungskonstante, (Periodendauer $T = \frac{1}{\nu}$)

$\omega_0 = \sqrt{\omega^2 + \beta^2}$ - ungedämpfte Kreisfrequenz

Linienform: $\Delta\nu = 2\beta = \frac{\Delta\omega}{2\pi} = \frac{1}{\tau}$ - Halbwertsbreite (Frequenz $\nu = \frac{c}{\lambda}$; Lebensdauer)

Oszillatorstärke (Wellenzahl $\tilde{\nu} = \frac{1}{\lambda}$)

$f = 4{,}3 \cdot 10^{-9}\, M \cdot cm^3 \int \varepsilon(\tilde{\nu})\, d\tilde{\nu} \to$ Gaußkurve $f = 4{,}3 \cdot 10^{-9} M \cdot cm^3 \frac{\varepsilon(max) \cdot \Delta\tilde{\nu}(FWHM) \sqrt{\pi}}{2\sqrt{\ln(2)}}$

$\to$ QM $f = \frac{8\pi^2 m_e \nu}{3he^2} |\mu_{if}|^2$ $|\mu_{if}|^2 = \frac{3e^2h}{8\pi^2 m_e \nu} f_{if}$ Übergangsdipolmoment $\omega_{max} = \frac{2\pi c}{\lambda} = 2\pi f = 2\pi c \tilde{\nu}$

strahlende Ratenkonstante $k_r = \frac{1}{4\pi\varepsilon_0} \frac{2}{\hbar} \left(\frac{\omega_{max}}{c}\right)^3 |\mu_{if}|^2$

Einstein-Koeffizienten

Absorption: $B_{12} \cdot N_1 \cdot \beta$ (Ratenkonstante; spektrale Energiedichte)

stim. Emission $B_{21} \cdot N_2 \cdot \beta$ $\Rightarrow \frac{dN_1}{dt} = -\frac{dN_2}{dt} = -B_{12} N_1 u(\nu) + B_{21} N_2 u(\nu) + A N_2$

spont. Emission $A \cdot N_2$ (Anzahl Quantensysteme in Zustand) ohne spont. Emission $\to B_{12} N_1 u(\nu) = B_{21} N_2 u(\nu)$

$\Rightarrow \frac{A}{B} = \frac{8\pi h\nu^3}{c^3}$ mit $B_{12} = B_{21}$

Linienbreite

$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_2'}$ $\Delta\nu(FWHM) = \frac{1}{2\pi T_1} + \frac{1}{\pi T_2'} \to \frac{1}{\pi T_2'} = \frac{1}{\pi T_2'(T\to\infty)} e^{-\frac{\Delta E}{k_B T}}$

(Abfallrate Magnetisierung in x-y-Ebene; Rate E die von Kernspinsystem an benachbarte Mol. abgegeben wird; Relaxationszeit durch Magnetfeldinhomogenitäten; bei niedrigem T vernachlässigbar; $T_2' = (\pi(\Delta\nu_{FWHM} - \frac{1}{2\pi T_1}))^{-1}$; $\tau_1 = \frac{1}{2\pi \Delta\nu_{FWHM}}$)

Dopplerverbreiterung

$\nu_B = \nu_0 \left(1 \pm \frac{v}{c}\right)$ (Bew. auf Sender zu / Bew. von Sender weg) $\Delta\nu = \frac{\nu_0}{c}\sqrt{\frac{8k_BT\ln(2)}{m}}$ $|v| = \left(\frac{\nu_B}{\nu_0} - 1\right)c = \left(\frac{\lambda_0}{\lambda_B} - 1\right)c$ $\Delta\lambda = \frac{\lambda_0}{c}\sqrt{\frac{8k_BT\ln(2)}{m}}$

VL 3 - Rotationsspektroskopie

$\nu_{H_\alpha}(\text{Erde/Stern}) = \frac{c}{\lambda_{H_\alpha}(\text{Erde/Stern})}$

$\to v = \frac{\nu_{H_\alpha}(\text{Erde})}{\nu_{H_\alpha}(\text{Stern})} - 1) c$ $\Delta\nu = \frac{\nu_{H_\alpha}(\text{Stern})}{c}\sqrt{\frac{8k_BT\ln(2)}{m}}$ $\to T = \frac{\Delta\nu^2 c^2 m}{8k_B \ln(2) \nu_{H_\alpha}^2(\text{Stern})}$

Freiheitsgrade	Linear	nicht-Linear
Translation	3FG	3FG
Rotation	2FG	3FG
Schwingung	3N-5FG	3N-6FG

Wärmekapazität $C_V = \frac{1}{2}(\nu_{Trans} + \nu_{Rot} + \nu_{Schwing}) R$

Starrer Rotator (Rotationsquantenzahl; Trägheitsmoment $I = \mu r^2$)

Rotationsenergie $E_{rot} = \frac{1}{2I} L^2 / E_{rot} = \frac{1}{2} I \omega^2$

Energieeigenwerte $E_J = \frac{\hbar^2}{2I} J(J+1)$ (Rotationsquantenzahl) $E_J = Bhc \cdot J(J+1)$

Auswahl $\Delta\mu_{if} = 0$, $\Delta J = \pm 1$ $|\mu_{J,J+1}|^2 = \mu_e^2 \frac{J+1}{2J+1} \Rightarrow \frac{N_J}{N_0} = \frac{J+1}{2J+1}$

Rotationskonstante aus Linienabstand $B = \frac{h}{8\pi^2 c I}$

Entartung $g_J = 2J+1$ H_2O $r = \sin\left(\frac{\alpha}{2}\right)$ $I = \left(2\frac{m_H}{N_A}\right) r_H^2 + \frac{m_O}{N_A} r_O^2$

Abstand zw. Niveaus $\Delta E = 2Bhc(J+1) / \Delta E = \frac{\hbar^2}{I}(J+1)$

Linienabstand $\Delta x = 2B = const.$ (aus hier gültig)

Rotationsspektroskopie

Voraussetzung: $\Delta\mu \neq 0$, fr. Rot.

Rotationsramanspektroskopie

Voraussetzung: $\Delta\alpha \neq 0$, Mol. schw.

Auswahl $\Delta J = \pm 1$ $J \to J+2$ Stokes $J \to J-2$ Anti-Stokes

Energiedifferenz $\Delta E = Bhc(4J+6)$

Linienabstand $\Delta x = 4B$

Nicht-starrer Rotator $r\uparrow, I\uparrow, B\downarrow$

Kreiseltypen

linear $I_a = 0, I_b = I_c$ oblat $I_a = I_b < I_c$

sphärisch $I_a = I_b = I_c$ asymmetrisch $I_a < I_b < I_c$

prolat $I_a < I_b = I_c$

$\hookrightarrow$ versch. Rot. konst. $A = \frac{h}{8\pi^2 c I_a}$ $B = \frac{h}{8\pi^2 c I_b}$ $C = \frac{h}{8\pi^2 c I_c}$

VL 4 - Schwingungsspektroskopie

Harm. Oszillator $\Delta v = \pm 1$ $k = \omega^2 m$ $\omega = \sqrt{\frac{k}{m}} = 2\pi c \tilde{\nu}$

Pot. Energie $V(x) = \frac{1}{2} k x^2$

Nullpunktsenergie $E_0 = \frac{1}{2}\hbar\omega$ (Quantenzahl)

Bindungsabstand $\Delta x^2 = \frac{\hbar^2}{\mu\omega}\left(v + \frac{1}{2}\right)$

Energieniveaus $\Delta E_v = \hbar\omega = h\nu / \Delta E = \hbar\omega\left(v + \frac{1}{2}\right)$

IR $\Delta v = \pm 1, \Delta J = \pm 1$

Raman $\Delta\alpha \neq 0, \Delta J = \pm 2$

Anharm. Oszillator $a = \sqrt{\frac{\mu}{2D_e}}\,\omega$

Pot. Energie $V(x) = D_e(x)(1 - e^{-ax})^2$ $D_e = hc\tilde{\nu}$; $\hookrightarrow D_0 = D_e - E_0$, $E_0 = \frac{1}{2}\hbar\omega$

Energieniveaus $\Delta E = \hbar\omega(1 - 2x_e(n+1))$ $x_e = \frac{h\omega_e}{8\pi D_e}$

Energieeigenwerte $E_n = \frac{h\omega_e}{2\pi}\left(n + \frac{1}{2}\right) - x_e \frac{h\omega_e}{2\pi}\left(n + \frac{1}{2}\right)$

VL 5 - Elektronische Anregung

Teilchen im Kasten

Energiewerte $E_n = \frac{n^2 h^2}{8 m_e a^2}$ Übergangsenergie $\Delta E = \frac{h^2}{8 m_e a^2}\left[\left(\frac{Q}{2}+1\right)^2 - \left(\frac{Q}{2}\right)^2\right]$ (Kastenbreite; Anzahl π-e⁻)

Franck-Condon-Übergänge $FC_{v,0} = \left(\frac{\mu\omega r^2}{2\hbar}\right)^v \frac{1}{v!} \exp\left(-\frac{\mu\omega r^2}{2\hbar}\right)$

VL 6 - Fluoreszenz

Dynamische Stokes-Verschiebung $\Delta E_{Stokes} = 2\Delta E_{VR} + 2\Delta E_{Solv}$ (Schwingungsrelax., Solvensrelax.)

Excitonen Potential Dipol-Dipol-WW $V = \frac{\mu_1\mu_2}{4\pi\varepsilon_0 r^3}(1 - \cos^2\theta)$

VL 7 - Löschung & FRET

Fluoreszenzausbeute $\Phi_F = \frac{\text{em. Photonen}}{\text{abs. Photonen}} = \frac{k_F}{k_F + k_{IC} + k_{ISC}} = \frac{k_F}{k_0} = k_F \cdot \tau_0$ (Fluoreszenzlebensdauer; $\tau_0 = \frac{1}{k_0}$)

Stern-Volmer-Gleichung $\frac{\Phi_{F_0}}{\Phi_{F_Q}} = \frac{\tau_0}{\tau_Q} = 1 + \frac{k_Q[Q]}{k_0} = 1 + \tau_0 k_Q [Q] = 1 + k_{SV}[Q]$ (Löschungskonst., Stern-Volmer-Konst., Quencher Konz.)

Fluoreszenzlebensdauer $\tau = \frac{1}{k_0 + k_Q[Q]}$ $\xrightarrow{\text{ohne } k_0}$ $\tau = \frac{\tau_0}{1 + \tau_0 k_Q [Q]}$

Smoluchowski-Gleichung $k_Q^{diff} = 4\pi R_0 D_0 N_A \Rightarrow k_Q^{diff} = \frac{k_B T}{1{,}5\eta} N_A R_0 \left(\frac{1}{R_a} + \frac{1}{R_b}\right) = \frac{4k_BTN_A}{1{,}5\eta} \frac{(1+a)^2}{a}$

Annihilation $S_1 + S_1 \to S_0 + S_1 + \Delta$, $T_1 + T_1 \to S_1 + S_0$

FRET

Effizienz $E = \frac{F_A}{F_A + F_D} = \frac{k_{FRET}}{k_{FRET} + k_F + k_{IC} + k_{ISC}} = \frac{R_0^6}{R_0^6 + R_{DA}^6} \Rightarrow R_{DA} = \sqrt[6]{\frac{R_0^6}{E} - R_0^6}$

$R_0 = \sqrt[6]{\frac{9\ln(10)}{128\pi^5 N_A} \frac{J\kappa^2\Phi_{F_D}}{n^4}}$ $J = \int_0^\infty f(\lambda)_D\, \varepsilon(\lambda)_A \lambda^4 d\lambda \to f(\lambda)_D$: Int ges. Messbereich von Fl.

weitere Effizienzen $E = 1 - \frac{I_{D(A)}}{I_{D(0)}}$ / $E = 1 - \frac{\tau_{D(A)}}{\tau_{D(0)}}$ (Anwesenheit Akzeptor / Abwesenheit)

VL 8 - Anisotropie

$r = \frac{F_{\parallel} - F_{\perp}}{F_{\parallel} + 2F_{\perp}}$ $\left(= \frac{F_{\parallel} - GF_{\perp}}{F_{\parallel} + 2GF_{\perp}},\ G = \frac{g_V}{g_H} = \frac{F_{\perp}}{F_{\parallel}}\right)$

Fundamentale Anisotropie $r(t) = r_0 e^{-\frac{t}{\tau_{rot}}}$

$\hookrightarrow r_0 = \frac{2}{5} \frac{3\cos^2(\beta) - 1}{2}$, $\beta = \cos^{-1}\sqrt{\frac{5r_0 + 1}{3}}$ (Rotationskorr. zeit; Fluoreszenzlebensdauer)

Perrin-Gleichung $\frac{r_0}{r} = 1 + \frac{\tau}{\beta} \to r = \frac{r_0}{1 + \frac{\tau}{\beta}} \to \beta = \frac{\tau}{1 + \frac{r_0}{r}}$

$\beta = -\frac{t}{\ln\left(\frac{r(t)}{r_0}\right)} \to \beta = \frac{\eta V}{k_B T} = \frac{4\pi\eta R^3}{3k_BT} = \frac{1}{6D} \to R = \sqrt[3]{\frac{3\beta k_B T}{4\pi\eta}}$

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mittl. Geschw. $\bar{v} = \sqrt{\frac{8RT}{\pi M}}$ quadr. gem. Geschw. $v = \sqrt{\frac{3RT}{M}}$

wahrsch. Geschw. $v^* = \sqrt{\frac{2RT}{M}}$ Median $\int_0^{v_{med}} \rho(v)\,dv = \frac{1}{2}$ $\langle v^3\rangle - 3\langle v\rangle\langle v^2\rangle + 2\langle v\rangle^3$

Streuung $\sigma = \sqrt{\langle v^2\rangle - \langle v\rangle^2}$ Varianz $S = \sigma^2$ Schiefe $\gamma_1 = \frac{\mu}{\sigma}$

Mittelwert $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ Standardabweichung $S_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

Pumprate $E_{ph} = \frac{hc}{\lambda}$ $N_{ph} = \frac{E_{Puls}}{E_{ph}}$ $\phi = \frac{N_{ph}}{\tau_{Puls}}$

$I = \frac{\phi}{A}$ $\omega = BI$ t_{eff} $\frac{1}{t_{eff}} = \frac{1}{\tau_{Puls}} + \omega$

Lebensdauer $\tau_{S_1 - S_0} = \frac{1}{k}$ $k = B_{S_1 - [on]} I$

$I = \dot{n}\frac{h\nu}{A}$ $\dot{n} = \frac{E}{\Delta t}$

Maxwell-Boltzmann $f(\nu) = \frac{m}{2\pi k_B T} e^{-\left(\frac{\nu}{\nu_P}\right)^2}$ $\nu_P = \sqrt{\frac{2k_BT}{m}}$

$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $\sigma = \frac{k_B T \nu_0^2}{c^2 m}$

Innere Konversion $\Delta E = E_{an} - \Phi_F E_F = \frac{hc}{\lambda_{an}} - \Phi_F \frac{hc}{\lambda_F} = hc\left(\frac{1}{\lambda_{an}} - \frac{1}{\lambda_F}\right)$

$\Delta E_L = \Delta T c_p \beta = Q$ $c = \frac{\frac{N}{L}}{N_A} = \frac{\frac{\Delta E_L}{\Delta E}}{\frac{N}{N_A}} = \frac{\Delta E_L}{N_A \cdot \Delta E}$

Franck-Condon $S(0,0) = \int_0^\infty \psi_0 \psi_0'\, dR = \frac{1}{\pi a^2}\int_0^\infty e^{\left(\frac{(R - R_e)^2}{2a^2} + \frac{(R - R_e')^2}{2a^2}\right)} dR$

$= \frac{1}{\pi a^2}\left[(R_e - R_e')^2 + 4(R^2 - R(R_e - R_e')\right]$

$a = R\frac{1}{2}(R_e - R_e')$ $S(0,0) = \exp\left(-\frac{(R_e - R_e')^2}{4a^2}\right)$

Dipolmoment $\mu = \sqrt{\mu_x^2 + \mu_y^2 + \mu_z^2}$

Stern-Volmer $\frac{d[A]}{[A]} = -k\,dt$ $\int_{[A]_0}^{[A]_t} \frac{1}{[A]}\, d[A] = \int_0^t -k\, dt$

$\hookrightarrow \ln\frac{[A]_t}{[A]_0} = -kt$ $[A]_t = [A]_0 e^{-kt}$

Kinetik Energietransfer $E = \frac{k_F}{k_F + k_D} \to k_D = k_F(E^{-1} - 1) \to k_F = \frac{\frac{1}{\tau_D}}{\frac{1}{E} - 1}$

Transporteigenschaften $\beta \ll \tau \to r \to 0 \to$ Anisotropie

$\beta \gg \tau \to r \to r_0 \to$ FCS

$D = \frac{\omega_0^2}{4\tau_D}$ $r = \frac{k_B T}{6\pi D\eta}$

	NMR	ESR	MW	IR	UV/VIS	X-Ray
λ	100-10m	1m-1cm	1cm-100µm	100-1µm	1µ-10nm	10nm-10pm
ν	10^6-$10^8\,s^{-1}$	10^8-$10^{10}\,s^{-1}$	10^{10}-$10^{12}\,s^{-1}$	10^{12}-$10^{14}\,s^{-1}$	10^{14}-$10^{16}\,s^{-1}$	10^{16}-$10^{18}\,s^{-1}$

SMKS - 2

Grenzflächen

$dA = 2b\,dx \quad dW = \gamma\, dA \quad \Delta W = \gamma \Delta A \quad F = -\frac{dW}{dx} = -2\gamma b \quad \gamma = \frac{F}{2l} = \left(\frac{\delta F}{\delta A}\right)_{T,V} = \left(\frac{\delta G}{\delta A}\right)_{p,T}$

Helmholtz-Energie $dF = \gamma\, dA$

Adhäsion: $W_{AB} = \underbrace{\gamma_{AE}\Delta A + \gamma_{BE}\Delta A}_{\text{Bildung neue Grenzfläche}} - \underbrace{\gamma_{AB}\Delta A}_{\text{Auflösung alte Grenzfläche}} = (\gamma_{AE} + \gamma_{BE} - \gamma_{AB})\Delta A$

Kohäsion: $W_{AA} = \gamma_{AE} \cdot 2\Delta A$

Benetzung 1) Tropfen bildet Linse: $S_{AB} > 0$, $W_{AA} > W_{AB}$ 2) Tropfen benetzt: $S_{AB} < 0$ 3) Film: $S_{AB} \approx 0$

Young-Gleichung $\gamma_{sg} = \gamma_{sl} + \gamma_{lg} \cdot \cos\theta$

Wenzel-Gleichung $\cos\theta_{app} = R_{rau} \cdot \cos\theta_{flach}$

Optik

Snellius $\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1}$

Kritischer Winkel $\theta_C = \sin^{-1}\left(\frac{n_2}{n_1}\right) = 41°$ $\theta_C = 90°$ Totalreflexion bei $\theta > \theta_C$, $n_2 > n_1$ Brechung zum Lot

Linsenschleifer-Formel $D = \frac{1}{f} = \frac{n_L - n}{n}\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Linsengleichung $\frac{1}{b} + \frac{1}{g} = \frac{1}{f}$ $\frac{B}{G} = \frac{b-f}{f} = \frac{b}{g}$

Numerische Apertur $NA = n \cdot \sin\alpha$

Abbe-Limit $d = \frac{\lambda}{2(n \cdot \sin\alpha)} = \frac{\lambda}{2NA}$

de-Broglie $\lambda = \frac{h}{p} = \frac{h}{m_e v}$ $E_{kin} = \frac{1}{2} m_e v^2 = e_0 U$ $\lambda = \frac{h}{\sqrt{2 m_e U}}$

Elektrostat. & magnet. Linsen $|F_{Lorentz}| = |q||\vec{v}||\vec{B}| \sin\alpha$

B-Feld: $B \sim \frac{nI}{D}$ – Strom – Anzahl Windungen – Durchmesser Spalte

Photoelektronenspektroskopie $E_{kin} = E_{Ph} - E_B\, \Phi_{spektro}$

Rastertunnelmikroskopie $\frac{d^2\psi(z)}{dz^2} = \frac{2m}{\hbar^2}(V-E)\psi(z)$ $\psi(z) = Ae^{-kz} + Be^{-kz}$ $k = \sqrt{\frac{2m(V-E)}{\hbar^2}}$

↳ vor Barriere $\psi_1(z) = Ae^{ik_1 z} + Be^{-ik_1 z}$ $k_1 = -ik = \sqrt{\frac{2mE}{\hbar^2}}$ Sonde: $\psi(z) = Fe^{ik_1 z} + Ge^{-k_1 z}$

↳ in Barriere $\psi_2(z) = De^{-k_2 z}$ $k = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$ $T = \frac{|F|^2}{|A|^2} = \frac{16E(V_0 - E)}{V_0^2} e^{-2k_2 d}$

Rasterkraftmikroskopie

Lennard-Jones-Potential: $V(d) = 4\varepsilon\left(\left(\frac{\sigma}{d}\right)^{12} - \left(\frac{\sigma}{d}\right)^{6}\right)$ Größe Objekt Durchmesser

Hooke'sches Gesetz $F = -kd$

Laser Besetzung: $\frac{N_1}{N_2} = \frac{g_1}{g_2} e^{-\frac{\Delta E}{k_B T}}$

Elektronenmikroskopie $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{m_e\sqrt{\frac{2E_{kin}}{m_e}}} = \frac{h}{\sqrt{2m_e E_{kin}}} = \frac{h}{\sqrt{2m_e eU}} = \frac{1{,}23 \cdot 10^{-9}\, m\, V^{1/2}}{\sqrt{U}}$

Präfixe

da - 10^1, h - 10^2, k - 10^3, M - 10^6 G - 10^9 T - 10^{12} Å - 10^{-10} m

d - 10^{-1}, c - 10^{-2}, m - 10^{-3}, µ - 10^{-6}, n - 10^{-12}, p - 10^{-15}, a - 10^{-18}

$1\,mL \mathrel{\hat=} 1\,cm^3 = 10^{-6}\,m^3$ $1\,cm^{-1} = 100\,m^{-1}$

$1\,L \mathrel{\hat=} 10^{-3}\,m^3 \Leftrightarrow 1\,m^3 \mathrel{\hat=} 1000\,L$

Einheiten

$1\,Hz = 1\,s^{-1}$ $1\,N = 1\,J \cdot m^{-1} = \frac{kg\,m}{s^2}$ $1\,Pa = \frac{N}{m^2} = \frac{kg}{m \cdot s^2}$ $J = 1\,Nm = \frac{kg\,m^2}{s^2}$ $1\,W = 1\,\frac{J}{s} = \frac{kg\,m^2}{s^3}$ $1\,C = 1\,As$

$1\,\Omega = 1\,\frac{V}{A} = 1\,\frac{kg\,m^2}{s^3 A^2}$ $1\,T = 1\,\frac{kg}{s^2 \cdot A}$ $1\,V = 1\,\frac{W}{A} = 1\,\frac{J}{C} = \frac{kg \cdot m^2}{s^3 \cdot A}$ $1\,F = 1\,\frac{C}{V} = 1\,\frac{s^4 \cdot A^2}{kg \cdot m^2}$

Konstanten

$c = 2{,}998 \cdot 10^8\,\frac{m}{s}$ $R = 8{,}314\,\frac{J}{mol \cdot K}$ $e = 1{,}602 \cdot 10^{-19}\,C$ $m_p = 1{,}672 \cdot 10^{-27}\,kg$ $m_e = 9{,}109 \cdot 10^{-31}\,kg$ $m_n = 1{,}674 \cdot 10^{-27}\,kg$

$\varepsilon_0 = 8{,}85 \cdot 10^{-12}\,F \cdot m^{-1}$ $N_A = 6{,}022 \cdot 10^{23}\,\frac{1}{mol}$ $h = 6{,}626 \cdot 10^{-34}\,Js$ $\mu_0 = 1{,}25 \cdot 10^{-6}$ $k_B = 1{,}38 \cdot 10^{-23}\,\frac{J}{K}$ $u = 1{,}66 \cdot 10^{-27}\,kg$ $eV = 1{,}602 \cdot 10^{-19}\,J$

$\mu_B = \frac{e\hbar}{2m_e} = 9{,}27 \cdot 10^{-24}\,J \cdot T^{-1}$ $\mu_e = -9{,}28 \cdot 10^{-24}\,J\,T^{-1} = -1{,}001\,\mu_B$ $F = eN_A = 96485\,C\,mol^{-1}$